# thunderclap

*Sleep through the transfer. Wake up to victory.*

User Guide · version 0.1.0 · the `clap` command

Fast, encrypted file and folder transfer between Linux machines. Use it from the terminal when you want speed and simplicity, or from a live dashboard in your browser when you want to see what is happening.

Parallel TCP streams · AES-256-GCM / ChaCha20-Poly1305 · PAKE pairing codes · Adaptive zstd · Resumable · LAN discovery · CLI + web UI

For Linux · written in C · GPL-3.0-or-later

github.com/hjani-2003/thunderclap · October 2026

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# 1 Introduction

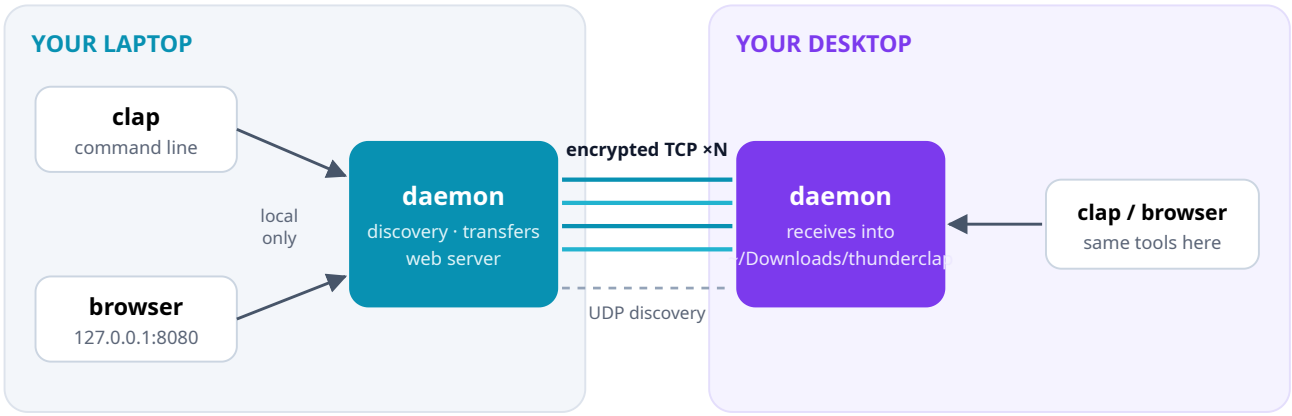
thunderclap moves files and folders between two Linux computers as fast as the connection between them allows: a single 1 GB video, a photo library, or a source tree with a hundred thousand tiny files.

> **THE NAME**
>
> thunderclap is named after *Thunder Breathing, First Form: Thunderclap and Flash* from Demon Slayer — one move, blindingly fast, performed in its user's sleep. The program is called **thunderclap**; the command you type is the short `clap`.

- **Fast**
  Splits data into 4 MiB chunks and sends them over several TCP connections at once (8 by default). Small files are packed together so thousands of them don't cost thousands of round trips.
- **Private**
  Everything is encrypted and authenticated. Machines prove who they are with keys exchanged once during pairing, so only your devices can send to you.
- **Smart compression**
  Text, logs and databases are compressed with zstd on the fly; videos, archives and ISOs are detected as incompressible and sent as-is at full speed.
- **Resumable**
  If the network drops, a machine crashes or you press Ctrl-C, just run the same command again — only the missing pieces are sent.
- **Zero setup on a LAN**
  thunderclap finds other thunderclap machines on your network automatically and shows which network interface and link speed each one is reachable on.
- **Two interfaces**
  A terminal command for speed and scripts, and a browser dashboard with a file picker, live throughput graphs and history.

## How it fits together

Each computer runs a small background program, the **daemon**. The daemons talk to each other over the network. You talk to *your* daemon either with the `clap` command or through the web dashboard — both are just remote controls for the same daemon. You never need to start the daemon by hand: any `clap` command starts it if it isn't running.




## Words used in this guide

| | |
|---|---|
| **Daemon** | The background program on each machine that does the actual work. Started automatically. |
| **Peer** | Another computer running thunderclap. |
| **Pairing** | A one-time introduction between two of your machines using a short code. Afterwards they recognise each other forever (until you unpair). |
| **One-shot code** | A code that lets an *unpaired* machine send you files exactly once — handy for a friend's laptop. |
| **Fingerprint** | A 16-character identifier of a machine's key, like `a786fdb2645b25f1`. Shown everywhere so you can tell devices apart. |
| **Transfer** | One send operation (any number of files and folders). Each has a small number, its *id*, used by commands like `clap cancel 3`. |
| **Download folder** | Where received files go: `~/Downloads/thunderclap` unless you change it. |

# 2 Installation

thunderclap is built from source. It takes about a minute and needs to be done on every machine you want to use.

## Requirements

- A Linux distribution (developed on Ubuntu; any modern distribution works).
- A C compiler with C11 support — any GCC or Clang from the last several years (tested with GCC 12 and 15).
- Four small libraries: **libsodium** 1.0.18 or newer (encryption), **zstd** (compression), **xxHash** (checksums) and **cJSON**.

## Build and install

terminal

```
# 1. Install the build tools and libraries (Debian / Ubuntu)
$ sudo apt install build-essential pkg-config libsodium-dev libzstd-dev libxxhash-dev
libcjson-dev

# 2. Get the code and build it
$ git clone https://github.com/hjani-2003/thunderclap.git
$ cd thunderclap
$ make

# 3. Install into ~/.local/bin (no root needed)
$ make install
installed to /home/you/.local/bin/clap
enable at login with: systemctl --user enable --now clap
```

> **FEDORA / ARCH**
> Package names differ slightly: Fedora uses `libsodium-devel libzstd-devel xxhash-devel cjson-devel`; Arch uses `libsodium zstd xxhash cjson`.

Make sure `~/.local/bin` is on your `PATH` (it is by default on Ubuntu after you log in again). Check the install:

terminal

```
$ clap --version
0.1.0
```

## Start thunderclap automatically (optional)

Any `clap` command starts the daemon when needed, but a machine can only *receive* while its daemon is running. On a computer that should always be ready to receive — a desktop or home server — let systemd start it when you log in:

terminal

```
$ systemctl --user enable --now clap
# optional: keep it running even when you are logged out
$ sudo loginctl enable-linger $USER
```

## Open the firewall

If a firewall is active (Ubuntu ships `ufw`), allow thunderclap's two ports on **every** machine that should receive or be discovered:

terminal

```
$ sudo ufw allow 7878/tcp   # transfers
$ sudo ufw allow 7879/udp   # finding each other on the LAN
```

## Uninstall

terminal

```
$ clap stop
$ systemctl --user disable clap; rm ~/.config/systemd/user/clap.service
$ rm ~/.local/bin/clap
$ rm -rf ~/.config/thunderclap      # removes keys, pairings and history
```

# 3 Quick start

Two machines on the same network: a LAPTOP that sends and a DESKTOP that receives. thunderclap is installed on both (chapter 2).

### 1 Check that the machines see each other LAPTOP

The first command starts the daemon. After a couple of seconds the other machine appears under "On this network".

laptop

```
$ clap peers
Paired:
  (none — pair with `clap pair`)

On this network:
  desktop              cafb151299672aaa  192.168.1.7:7878  via wlp1s0 (Wi-Fi)
```

Nothing listed? Run any thunderclap command on the desktop too (e.g. `clap status`) so its daemon starts, and check the firewall (chapter 2).

### 2 Show a pairing code DESKTOP

desktop

```
$ clap pair
Pairing code:  3704-9640-4439

On the other machine run:
  clap pair desktop 3704-9640-4439

(valid for 5 minutes, one attempt)
```

### 3 Enter the code LAPTOP

laptop

```
$ clap pair desktop 3704-9640-4439
✓ paired with desktop (cafb151299672aaa)
```

The desktop's terminal confirms with `✓ paired with laptop`. You never need to do this again for these two machines.

### 4 Send something LAPTOP

```
laptop

$ clap send ~/Videos/trip ~/project --to desktop
[████████████████████▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒]  54.2%  610 MiB / 1.1 GiB  8.15 Gbit/s  ETA 0:01
✓ sent to desktop (1.1 GiB, 1502 files) in 1.1s — 8.20 Gbit/s avg
```

The files appear on the desktop in `~/Downloads/thunderclap/trip` and `~/Downloads/thunderclap/project`, with their permissions and modification times preserved.

### 5 Open the dashboard EITHER

```
terminal

$ clap ui
```

Your browser opens on `http://127.0.0.1:8080` with live graphs, a file picker and history (chapter 6).

**BOTH DIRECTIONS**

Pairing works both ways. The desktop can now also run `clap send … --to laptop`.

# 4 Pairing and trust

thunderclap only accepts transfers from machines it knows. There are two ways to become known: pairing (permanent) and one-shot codes (single use).

## Pairing two of your machines

Run `clap pair` on one machine to show a code, then `clap pair <name> <code>` on the other, as in the quick start. Rules worth knowing:

- A code is valid for **5 minutes** and allows **one attempt**. A wrong entry uses it up — just run `clap pair` again.
- `<name>` can be the device name shown by `clap peers`, a fingerprint prefix, or an address like `192.168.1.20` or `myserver.example.com:7878`.
- Separators in the code don't matter: `3704 9640 4439` and `370496404439` work too.
- The code never crosses the network. Even someone recording your traffic cannot work it out afterwards (the technique is called a PAKE — see chapter 9).

## One-shot codes for machines you don't own

To receive files from a friend's computer without pairing it permanently, ask for a one-shot code on the receiving side:

desktop (receiver)

```
$ clap recv --code
One-shot code:  0271-5949-4950

On the sending machine run:
  clap send <files> --to desktop --code 0271-5949-4950

Waiting for the transfer… (Ctrl-C to stop waiting)
Receiving server.log from laptop
✓ received from laptop (300.0 MiB, 1 files) in 0.3s — 8.33 Gbit/s avg, compressed
  saved to ~/Downloads/thunderclap/server.log
```

friend's laptop (sender)

```
$ clap send server.log --to desktop --code 0271-5949-4950
✓ sent to desktop (300.0 MiB, 1 files) in 0.3s — 8.43 Gbit/s avg, compressed
```

The code is valid for **10 minutes**, for **one** transfer, and leaves no permanent trust behind. In the dashboard, the same thing is the *Receive with code* button.

## Sending over the internet

Automatic discovery only works on the local network, but pairing and transfers work anywhere:

1. On the receiver's router, forward **TCP port 7878** to the receiving computer.
2. Pair using the public address: `clap pair home.example.com:7878 <code>` (you can pair on the LAN first, then travel — the pairing stays valid).
3. Send with the address: `clap send big.iso --to home.example.com:7878`. For paired devices, thunderclap also remembers the last address it reached them on.

> **TIP**
> Over long distances, more parallel streams help: try `--streams 16`.

## Seeing and removing paired devices

laptop

```
$ clap peers
Paired:
  desktop              cafb151299672aaa  192.168.1.7:7880  online

On this network:
  desktop              cafb151299672aaa  192.168.1.7:7880  via wlp1s0 (Wi-Fi)  [paired]
$ clap unpair desktop
forgot desktop
```

Unpairing only changes the machine you run it on: the laptop now refuses transfers from the desktop and can no longer send to it, but the desktop still lists the laptop until you run `clap unpair laptop` there too.

# 5 Using the terminal

Everything thunderclap can do is available as a `clap` sub-command. This chapter covers daily use; chapter 12 lists every option.

## Sending

```
$ clap send <file or folder>... --to <peer> [options]
```

- Send any mix of files and folders in one go. Folders are sent with everything inside them, including empty folders and symbolic links.
- `--to` takes a device name, a fingerprint prefix, or `host[:port]`.
- The progress bar shows percentage, amount, current speed, time left and — when compression is helping — how much smaller the data got.
- Press `Ctrl-C` to cancel. Running the same command later **resumes** where it stopped (chapter 8).

| OPTION | MEANING |
|---|---|
| `--streams N` | Number of parallel connections (1–32). Default: the number of CPU cores, between 2 and 8. |
| `--compress MODE` | `auto` (default) compresses only data that benefits; `off` never compresses; `1`–`19` forces that zstd level (higher = smaller but slower). |
| `--code CODE` | Send to an unpaired machine using a one-shot code it gave you. |
| `--detach` | Start the transfer in the background and return immediately. |

## Receiving

On a paired machine there is nothing to do: transfers are accepted automatically and saved into the download folder. Each top-level file or folder you send becomes an entry there:

- Sending `~/Videos/trip` creates `~/Downloads/thunderclap/trip/…`
- If something with that name already exists, thunderclap never overwrites it — it picks a new name like `trip (1)` or `notes (1).txt`.
- While a file is arriving it is called `name.clap-part`; it gets its real name, permissions and timestamp once complete.

### Asking before accepting

If you'd rather approve each incoming transfer, turn off auto-accept on the receiving machine:

desktop

```
$ clap config auto_accept off
$ clap ls
  id  started         peer            what              size          speed       state
   1  10-25 19:17  ← laptop          holiday      400.0 MiB   0.0%  0 kbit/s    waiting
$ clap accept 1        # or: clap reject 1
✓ received from laptop (400.0 MiB, 1 files) in 1.4s — 8.99 Gbit/s avg
  saved to /home/you/Downloads/thunderclap/holiday
```

`clap accept` shows the progress bar until the transfer finishes. An unanswered request is declined after 10 minutes. The dashboard shows *Accept* / *Decline* buttons for waiting transfers.

## Following transfers

laptop

```
$ clap send ~/Videos --to desktop --detach
transfer 5 started — follow it with `clap watch 5`
$ clap ls
  id  started         peer            what                size          speed
state
   5  10-25 19:07  → desktop         Videos            1.1 GiB   5.8%  207.7 Mbit/s
active
   4  10-25 19:07  → desktop         link benchmark        0 B   0.0%  24.20 Gbit/s   done
   3  10-25 19:07  → desktop         server.log       300.0 MiB 100.0%  8.43 Gbit/s    done
$ clap watch 5          # live progress bar; Ctrl-C here cancels the transfer
$ clap cancel 5         # stop it without watching
$ clap history          # the last 200 transfers, kept across restarts
```

Arrows show direction: `→` sent, `←` received.

## Checking status and settings

laptop

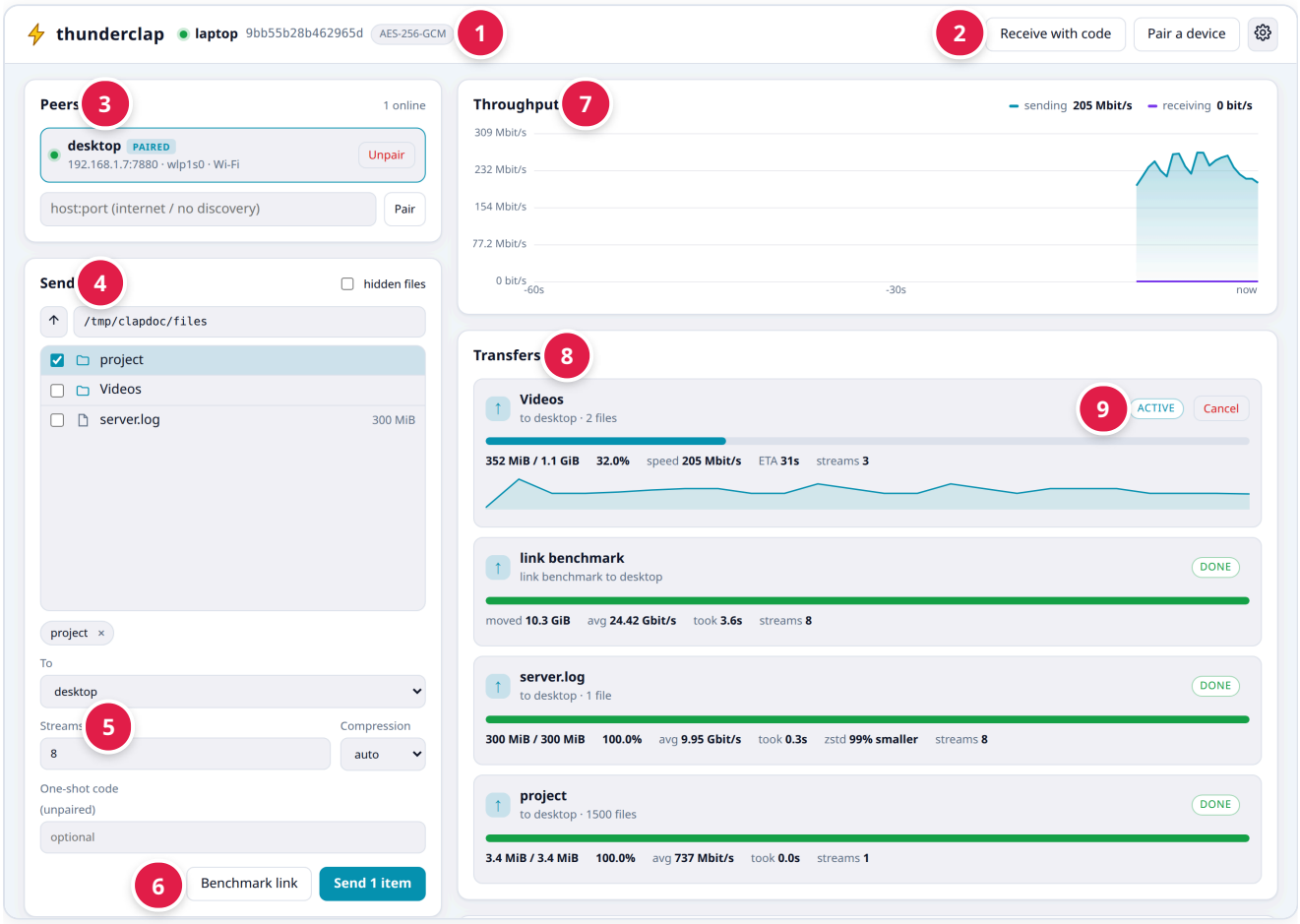
```
$ clap status
thunderclap 0.1.0 — daemon running
  device       laptop (a786fdb2645b25f1)
  peer port    7878 (TCP), discovery UDP 7879
  web UI       http://127.0.0.1:8080  (open with `clap ui`)
  downloads    /home/you/Downloads/thunderclap
  auto-accept  on from paired peers
  streams      8    compression auto    cipher AES-256-GCM
  interface    wlp1s0     192.168.1.7     Wi-Fi
$ clap config name "work-laptop"
$ clap config download_dir ~/Inbox
$ clap config compress off
$ clap stop             # stop the daemon (it restarts on the next thunderclap command)
```

# 6 Using the web dashboard

The dashboard does everything the terminal does, with a file picker and live visuals. It runs inside the daemon — there is no separate server to install.

## Opening it

Run `clap ui`. It opens your browser on `http://127.0.0.1:8080` with a secret access token attached to the address. The token keeps other websites you visit from controlling thunderclap, so open the dashboard with `clap ui` rather than typing the address. If you see “Open thunderclap from your terminal”, that is why. The dashboard is only reachable from the same computer.



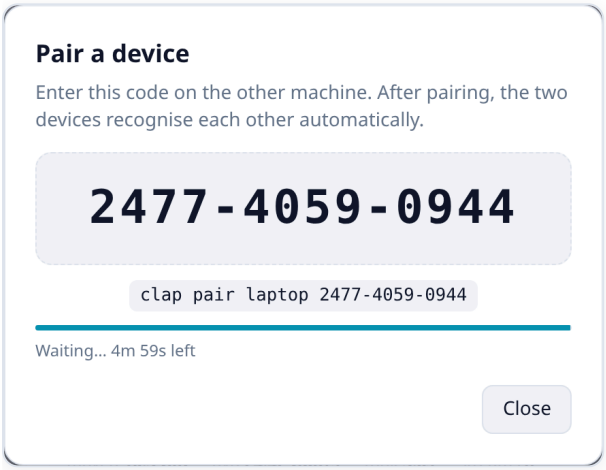

**The dashboard during a transfer.** Numbers refer to the list below.

1. **This device** — its name, fingerprint and the cipher in use. The dot is green while the page is connected to the daemon.
2. **Receive with code** and **Pair a device** show a code for the other machine; the gear opens Settings.
3. **Peers** — paired devices and new devices on the network, with address, interface and link speed. Click a paired device to make it the target; click *Pair* on a new one to enter its code. The box below pairs by address (internet).
4. **File picker** — type a folder path or use ↑. Click rows to select files and folders (double-click a folder to open it). Selections show as chips below.

5. **Options** — target device, number of streams, compression, and a one-shot code when sending to an unpaired device.
6. **Send** starts the transfer; **Benchmark link** runs a 10-second speed test to the target (chapter 7).
7. **Throughput** — total sending (cyan) and receiving (violet) speed over the last minute.
8. **Transfers** — one card per transfer with progress, speed, time left, compression savings and its own speed graph. Grey on the bar means data already present from an earlier attempt.
9. **State and actions** — Cancel, or Accept/Decline for incoming transfers waiting for approval. Errors appear in red under the card.

Below the transfers, **History** lists past transfers with size, average speed, time and result. Notifications pop up in the bottom-right corner when a transfer finishes or fails.

## Pairing from the browser

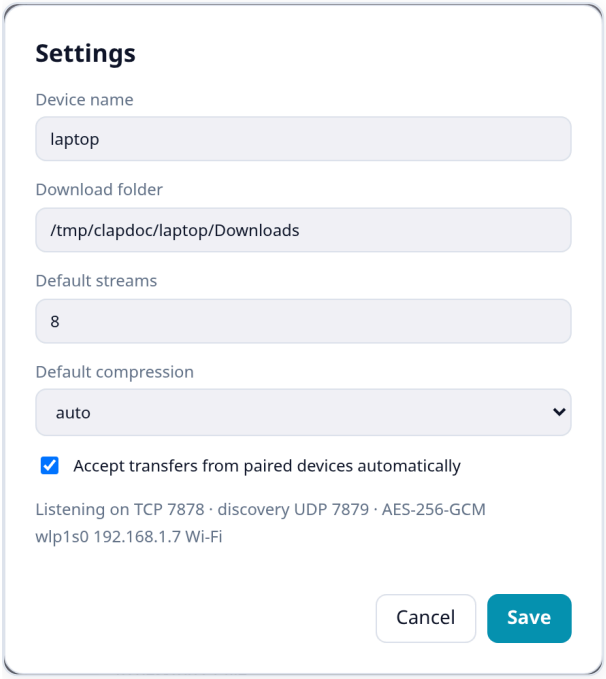



**Pair a device.** The code, the matching command for the other machine, and a countdown. The dialog closes by itself once pairing succeeds.

On the machine that *shows* the code, press **Pair a device**. On the other machine either run the displayed `clap pair …` command or click **Pair** next to the device in the Peers list and type the code.

**Receive with code** works the same way for one-shot transfers: give the code to the sender, who enters it in the *One-shot code* field (5) or uses `--code`.

# Settings

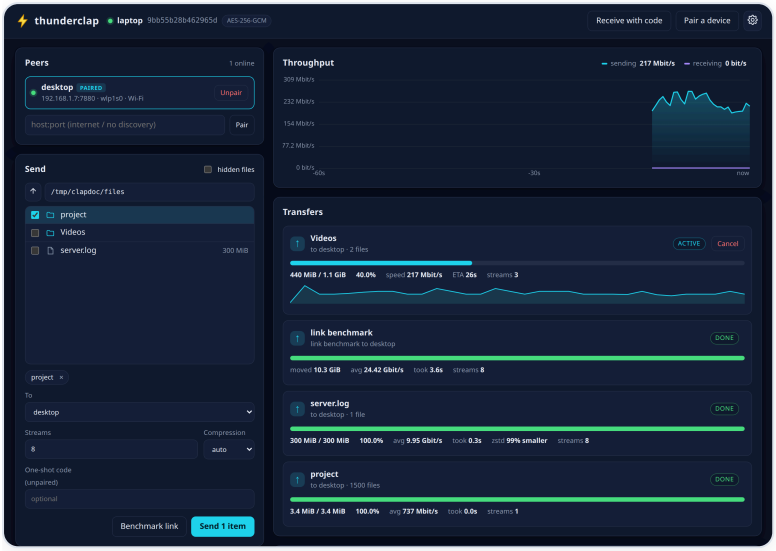



**Settings.** Stored in `~/.config/thunderclap/config.json`; the same as `clap config`.

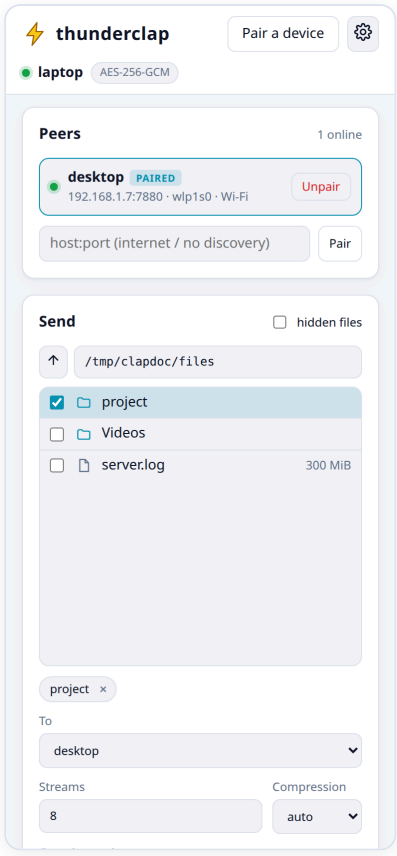
- **Device name** — how other machines see this one.
- **Download folder** — where received files go (an absolute path; created if missing).
- **Default streams** and **compression** — used when a send doesn't specify them.
- **Accept automatically** — turn off to approve each incoming transfer.

The bottom of the dialog lists the ports and the network interfaces with their link speed.

# Dark mode and phones




**Dark theme**, chosen automatically from your system setting.




**Narrow windows** stack the panels in one column.

# 7 Getting the most speed

A transfer can never be faster than the slowest of three things: the network link, the disks at either end, and the CPU doing encryption and compression. thunderclap is designed so the network or the disk is the limit.

## Measure your link first

`clap bench` sends encrypted data from memory to memory for a few seconds — no disks involved — and reports what the connection plus encryption can do:

laptop

```
$ clap bench desktop --secs 10
27.5 GiB  27.99 Gbit/s  0:01
link to desktop: 26.18 Gbit/s  (32.4 GiB in 10.6s over 8 streams)
```

This example ran with both daemons on one computer, so no real network was involved; between two machines the result is limited by the link (about 0.94 Gbit/s on gigabit Ethernet).

Compare the benchmark with a real transfer. If transfers are much slower than the benchmark, the disk (or many tiny files) is the bottleneck, not the network.

## What to expect

| CONNECTION | REALISTIC CEILING | 1 GB TAKES ABOUT |
|---|---|---|
| Wi-Fi 5 / Wi-Fi 6 | 200 – 900 Mbit/s, varies a lot | 10 – 40 s |
| Gigabit Ethernet | ~940 Mbit/s | ~9 s |
| 2.5 GbE | ~2.35 Gbit/s | ~3.5 s |
| 10 GbE | ~9.4 Gbit/s (fast NVMe needed) | ~1 s |
| USB4 / Thunderbolt networking | 10 – 30 Gbit/s | under 1 s |

For reference, on a single laptop (loopback, RAM disk) thunderclap measured about 13 Gbit/s for a 1 GiB file, 12 Gbit/s for a 20 000-file folder, and 22–25 Gbit/s in `clap bench`, with AES-256-GCM encryption on.

## Use the fastest cable you have

- **Prefer Ethernet to Wi-Fi.** When a peer is reachable over several interfaces, thunderclap automatically uses the one with the highest link speed — `clap peers` shows which one.
- **Direct cable.** Two machines connected by one Ethernet cable work fine; give each port an address (most desktops do this automatically with link-local addresses).

- **USB4 / Thunderbolt.** Connecting two machines with a Thunderbolt cable creates a `thunderbolt0` network interface. thunderclap uses it like any other network — this is the fastest option for laptops.
- **Bluetooth is not supported**: at about 2 Mbit/s in practice it is hundreds of times slower than Ethernet.

## Tuning knobs

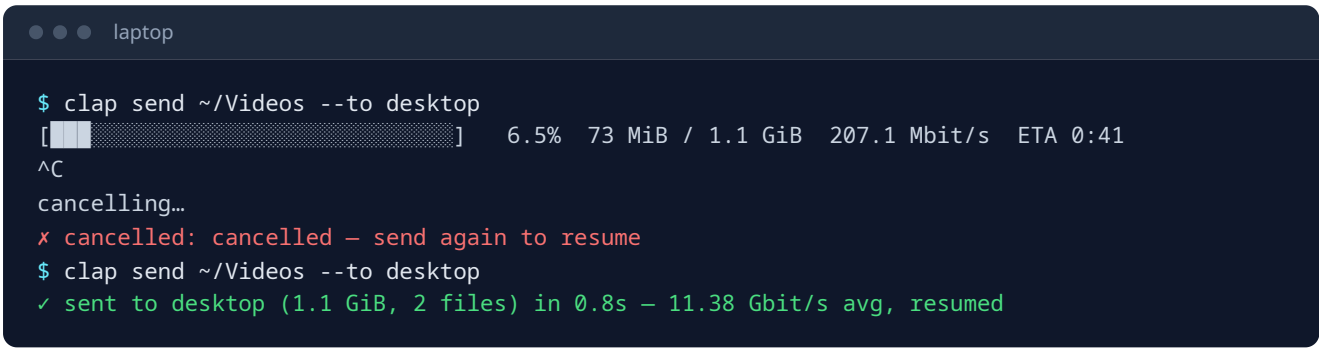
| SITUATION | TRY |
|---|---|
| Internet or long-distance link | `--streams 16` (or up to 32) to keep more data in flight. |
| Slow link, compressible data (logs, text, databases, VM images) | `--compress 3` to `9`: smaller on the wire. `auto` already uses a fast level. |
| Very fast link, data known to be incompressible | `--compress off` saves a little CPU (auto mode already skips incompressible chunks after a quick test). |
| Old CPU without AES-NI | Nothing to do: thunderclap switches to ChaCha20-Poly1305 automatically, which is fast in software. |
| Hundreds of thousands of small files | Already handled: files under 1 MiB are bundled into 4 MiB chunks. The receiving disk's speed at creating files becomes the limit. |

# 8 Reliability and resuming

Large transfers get interrupted: laptops sleep, Wi-Fi drops, someone presses Ctrl-C. thunderclap is built so that starting again costs only the part that didn't arrive.

## Resuming

To resume, **run the same send again** — same files, same destination device. thunderclap recognises it and transfers only the missing chunks:




This works after any kind of interruption — a cancel on either side, a lost connection, or a crash or reboot of either machine. Good to know:

- Resume works when the source files are **unchanged**. If a file was modified in the meantime, thunderclap treats it as a new transfer and saves it next to the old one (e.g. `Videos (1)`).
- The partly received data lives in the download folder as `*.clap-part` files plus a small progress journal in `~/Downloads/thunderclap/.clap/`. Deleting those simply means the next attempt starts from scratch; if you delete only some of the files, thunderclap notices and re-sends what is missing.
- Sending the same files again while that transfer is still running is refused with “this transfer is already running”.

## How thunderclap knows the data is correct

- Every record on the wire carries an authentication tag; any corruption or tampering in transit is detected and the connection is dropped.
- Every chunk carries an XXH3 checksum of its original content, checked after decompression and before it is written. A chunk that fails is simply requested again.
- At the end, the receiver confirms that every chunk is on disk before the sender reports success. Progress is only recorded after the data has been flushed to disk, so a power cut cannot leave thunderclap believing it has data it doesn't.
- Files keep their permissions, modification times and symbolic links; empty files and folders are recreated.

## When something goes wrong

Errors are reported on both sides in plain language, for example:

```
✗ failed: receiver: cannot write /home/you/Downloads/thunderclap/Videos/day1.mp4: No space left on device
```

Fix the cause (here: free some space) and run the same send again — it continues where it stopped.

# 9 Security

This chapter explains what protects your data, for readers who want to know. You don't need it to use thunderclap.

## Device identity

On first start every machine creates a random key pair (X25519), stored in `~/.config/thunderclap/identity.key` and readable only by you. Its fingerprint is what you see next to the device name. Pairing exchanges these public keys; afterwards each machine only accepts transfer sessions from keys in its paired list.

## Pairing codes cannot be cracked offline

Pairing uses **CPace**, a password-authenticated key exchange (PAKE) over the ristretto255 group. Both machines turn the code into a mathematical starting point and exchange values derived from it. If both used the same code they arrive at the same secret key; if not, they don't — and the exchanged values reveal nothing that would let an eavesdropper test guesses later. An attacker gets exactly one online guess per code, and a failed guess burns the code.

## Every transfer gets fresh keys

Each transfer between paired machines starts with a handshake modelled on the Noise “KK” pattern: both sides combine their permanent keys with brand-new temporary keys. Only the two genuine devices can compute the result, and recorded traffic stays unreadable even if a device key is stolen later (forward secrecy). Each parallel data connection derives its own keys, salted with a random value from the receiver, so recorded connections can't be replayed.

## Encryption

All traffic after the handshake is encrypted and authenticated with **AES-256-GCM** (on CPUs with AES instructions) or **ChaCha20-Poly1305**, using libsodium. File names and folder structure are encrypted too — someone watching the network sees only that two machines are talking, and roughly how much data moves. Separately, on your local network each daemon announces its device name and fingerprint every two seconds so others can find it; choose a device name you don't mind neighbours on the same network seeing.

## The receiving machine is protected

- Files can only be written inside the download folder: paths that are absolute, contain `..`, or would pass through a symbolic link created by the same transfer are rejected.
- Existing files are never overwritten.
- Unpaired machines are refused unless you armed a one-shot code yourself.

## The dashboard

The web server listens on `127.0.0.1` only. Every request must carry the token from `~/.config/thunderclap/web_token`, and requests whose `Host` or `Origin` isn't the local dashboard are rejected — this blocks malicious websites from using your browser to control thunderclap (CSRF and DNS rebinding). The terminal talks to the daemon over a Unix socket that only your user can open.

> **KEEP IN MIND**
>
> Anyone who can log in as your user can use thunderclap as you, like any other program. Treat `~/.config/thunderclap` as private, and unpair devices you no longer own.

# 10 Configuration reference

## Settings

Change settings with `clap config <key> <value>`, in the dashboard's Settings dialog, or by editing `~/.config/thunderclap/config.json` (restart the daemon afterwards with `clap stop`).

| KEY | DEFAULT | MEANING |
|---|---|---|
| `name` | host name | Device name shown to other machines. |
| `download_dir` | `~/Downloads/thunderclap` | Where received files are saved. |
| `auto_accept` | `on` | Accept transfers from paired devices without asking. |
| `streams` | CPU cores, 2–8 | Default number of parallel connections. |
| `compress` | `auto` | `auto`, `off`, or a zstd level `1`–`19`. |
| `peer_port` | `7878` | TCP port for transfers (config file or `clap daemon --port`). |
| `web_port` | `8080` | Dashboard port on 127.0.0.1 (config file or `clap daemon --web`; `0` disables it). |

## Files

| PATH | CONTENTS |
|---|---|
| `~/.config/thunderclap/config.json` | The settings above. |
| `~/.config/thunderclap/identity.key` | This device's private key. Deleting it gives the device a new identity — all pairings must then be redone. |
| `~/.config/thunderclap/peers.json` | Paired devices: name, public key, last known address. |
| `~/.config/thunderclap/web_token` | Access token for the dashboard. |
| `~/.config/thunderclap/history.json` | The last 200 transfers. |
| `~/.config/thunderclap/daemon.log` | Daemon log when it was started automatically (with systemd: `journalctl --user -u clap`). |
| `<download_dir>/.clap/` | Progress journals of unfinished incoming transfers. |
| `$XDG_RUNTIME_DIR/clap.sock` | Socket the `clap` command uses to talk to the daemon. |

## Network ports

| PORT | USED FOR | NEEDS TO BE OPEN? |
|---|---|---|
| `7878/tcp` | Pairing and transfers | Yes, on machines that receive (forward it on the router for internet use). |
| `7879/udp` | Discovery beacons (broadcast and multicast group 239.255.77.78) | Yes, for automatic discovery on the LAN. |
| `8080/tcp` | Dashboard, on 127.0.0.1 only | No — never reachable from the network. |

## Running several independent instances

`clap --home DIR …` (or the environment variable `CLAP_HOME=DIR`) uses a different configuration folder — and therefore a separate daemon, identity and pairing list. Give each instance its own ports:

```
$ clap --home ~/thunderclap-test daemon --port 7880 --web 8081 --name test-instance
```

This is how the automated tests run two "machines" on one computer.

# 11 Troubleshooting and FAQ

## Error messages

| MESSAGE | WHAT IT MEANS AND WHAT TO DO |
|---|---|
| nothing discovered yet — is thunderclap running on the other machine, and UDP 7879 open? | Start thunderclap on the other machine (`clap status`) and open UDP 7879. Some Wi-Fi networks (guest networks, “client isolation”) block discovery — use the address instead: `--to 192.168.1.20`. |
| X is not paired — run `clap pair` on it, or use a one-shot code | You found the device but haven't paired with it yet (chapter 4). |
| ... does not know this device — pair first with `clap pair` | The other machine doesn't (or no longer) have you in its paired list — e.g. it was reinstalled or unpaired you. Run `clap unpair` on your side and pair again. |
| ... has no active code — run `clap pair` there first | There is no code waiting on the other machine, it expired, or it was already used. Generate a new one. |
| wrong code (the code is now used up) | The code didn't match. Run `clap pair` (or `recv --code`) again for a fresh code. |
| no device called 'X' on this network. Found: ... | The name doesn't match any device. Long names copied from laptop model numbers are easy to mistype (`14IKB` vs `141KB`) — copy it from the list, use the first few characters of its fingerprint, or its IP address. Shorter names help: `clap config name lenovo` on that machine. |
| cannot reach HOST:7878 (No route to host / Connection timed out) | The device is visible (it appears in `clap peers`) but direct connections to it are blocked. Either a firewall on it (allow TCP 7878), or the Wi-Fi router's **client isolation / AP isolation**, which lets broadcasts through but stops devices from talking to each other. Turn isolation off in the router's wireless settings (don't use a guest network), connect one machine by cable, or use a phone hotspot. Test with `ping` to the other machine's address. |
| cannot connect to HOST:7878: Connection refused | The machine is reachable but thunderclap isn't running there, or listens on another port. |
| this transfer is already running | An identical transfer is still in progress. Wait for it, or cancel it first. |
| connection lost — run the same send again to resume | The network dropped or the other side stopped. Just repeat the command. |
| receiver: cannot write ...: No space left on device | The receiving disk is full (or the folder isn't writable). Fix it, then resend to resume. |

| MESSAGE | WHAT IT MEANS AND WHAT TO DO |
|---|---|
| cannot listen on port 7878: Address already in use | Another program (or another thunderclap) uses the port. Choose another with `clap daemon --port`. |
| “Open thunderclap from your terminal” in the browser | The page was opened without the access token. Run `clap ui`. |

## Frequently asked questions

### Where did my files go?

Into the receiver's download folder — `~/Downloads/thunderclap` by default. The final line of `clap recv` / `clap accept` prints the exact path, and the dashboard shows it on the transfer card.

### Can I send to several machines at once, or in both directions?

Yes. Start as many sends as you like (use `--detach` or several terminals); every daemon sends and receives at the same time.

### Is it safe to use over the internet?

Yes — everything is encrypted and authenticated, and an open port 7878 only accepts paired devices (or an armed one-shot code). As with any exposed service, keep thunderclap updated.

### Why is my transfer slower than the benchmark?

The disk is usually the limit: spinning hard drives manage 100–250 MB/s, and creating very many tiny files is slow on any disk. Wi-Fi speed also varies from second to second — look at the throughput graph.

### Does thunderclap sync folders like rsync?

No. It copies what you send. Sending a folder again creates a new copy ( `name (1)` ) unless the previous transfer of the same, unchanged files was unfinished — then it resumes.

### How do I see what the daemon is doing?

Read `~/.config/thunderclap/daemon.log`, or run it in the foreground with extra detail: `clap stop` then `clap daemon -v`.

### I reinstalled my OS. Do I need to pair again?

Yes, unless you kept `~/.config/thunderclap`. A fresh install creates a new identity; unpair the old entry on the other machines and pair again.

# 12 Command reference

`clap [--home DIR] <command> [options]` — `<peer>` is a device name, a fingerprint prefix, or `host[:port]`.

| COMMAND | DESCRIPTION |
|---|---|
| `send <paths>… --to <peer>` | Send files and folders. Options: `--streams N`, `--compress auto\|off\|1-19`, `--code CODE`, `--detach`. Ctrl-C cancels (resumable). |
| `recv --code` | Show a one-shot code (10 min, single use) and wait for the incoming transfer. |
| `pair` | Show a pairing code (5 min, one attempt) and wait until the other machine uses it. |
| `pair <peer> <code>` | Pair with the machine that shows the code. |
| `unpair <peer>` | Forget a paired device. |
| `peers` | Paired devices (online or not) and devices discovered on the network, with interface and link speed. |
| `ls` / `transfers` | Current and recent transfers with their ids. |
| `history` | The last 200 transfers, kept across restarts. |
| `watch <id>` | Show the live progress bar of a transfer. Ctrl-C cancels it. |
| `accept <id>` | Accept a waiting incoming transfer and follow it. |
| `reject <id>` | Decline a waiting incoming transfer. |
| `cancel <id>` | Stop a transfer (either direction). It can be resumed later. |
| `bench <peer> [--secs N] [--streams N]` | Measure link + encryption throughput without disk I/O (default 10 s, max 120 s). |
| `ui` | Open the dashboard in the browser (prints the address too). |
| `status` / `info` | Device, ports, download folder, defaults, cipher and network interfaces. |
| `config <key> <value>` | Change `name`, `download_dir`, `auto_accept` (on/off), `streams` or `compress`. |
| `daemon [options]` | Run the daemon in the foreground: `--port N`, `--web N`, `--dir DIR`, `--name NAME`, `-v` for debug logs. Options given here are saved. |

| COMMAND | DESCRIPTION |
| --- | --- |
| `stop` | Stop the daemon. |
| `--version`, `--help` | Print the version or the usage summary. |

## Exit status

`0` on success, `1` when a transfer or command fails, `2` for usage errors — useful in scripts:

```
$ clap send /srv/backups/nightly.tar.zst --to nas && echo "backup copied"
```

# 13 For developers

## Project layout

| PATH | CONTENTS |
|---|---|
| `src/main.c` | Command-line interface |
| `src/daemon.c`, `http.c`, `api.c` | Daemon loop and Unix-socket API, HTTP/SSE server, commands shared by CLI and dashboard |
| `src/peer.c`, `crypt.c` | Connection hellos, CPace pairing, KK handshake, encrypted channels |
| `src/sender.c`, `receiver.c` | The transfer engine |
| `src/manifest.c`, `journal.c`, `compress.c` | File lists and chunk layout, resume journal, adaptive zstd |
| `src/discovery.c`, `transfer.c`, `config.c` | LAN beacons, transfer state and history, settings and peer store |
| `src/proto.h` | Wire protocol definition |
| `web/` | Dashboard (plain HTML, CSS, JavaScript — no build step), embedded into the binary |
| `tools/bin2c.c` | Build helper that turns the dashboard files into C arrays |
| `tests/` | Unit tests and the end-to-end script |
| `docs/` | This guide: source in `docs/guide/`, screenshots in `docs/img/`, tools in `docs/tools/` |

## Tests

terminal

```
$ make test
== build/test_crypto
  ok
== build/test_storage
  ok
all unit tests passed
$ make e2e
== pairing
  ✓ wrong pairing code rejected
  ✓ paired alpha ↔ bravo
== transfers
  ✓ 300 MiB file
  ✓ folder: 3000 small files, modes, mtimes, symlink, empty file/dir
  ✓ name clash saved as 'big (1).bin'
  ✓ unpaired device refused
  ✓ one-shot code transfer from an unpaired device
== resume
  ✓ killed sender mid-transfer, resumed without resending finished chunks
== web
  ✓ web API requires the token and a loopback Host
== bench
  ✓ link benchmark
all 10 end-to-end checks passed
```

## Rebuilding this guide

The guide is an HTML page printed to PDF by headless Chrome (Node.js 22+ and Google Chrome or Chromium required):

```
$ make docs                # → docs/thunderclap-user-guide.pdf
# refresh the dashboard screenshots (needs a running daemon, ideally mid-transfer):
$ node docs/tools/screenshots.mjs "$(clap ui | head -1)" docs/img ~/some/folder
```